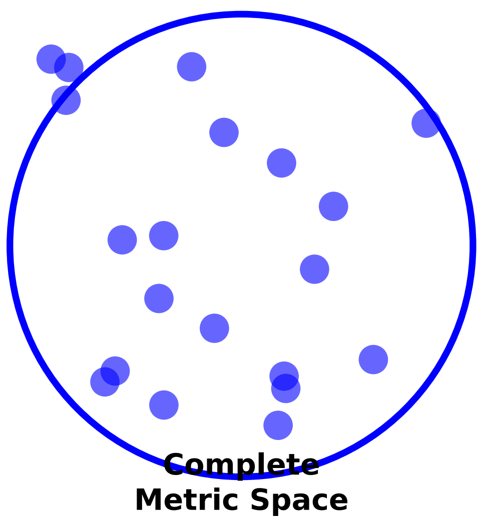
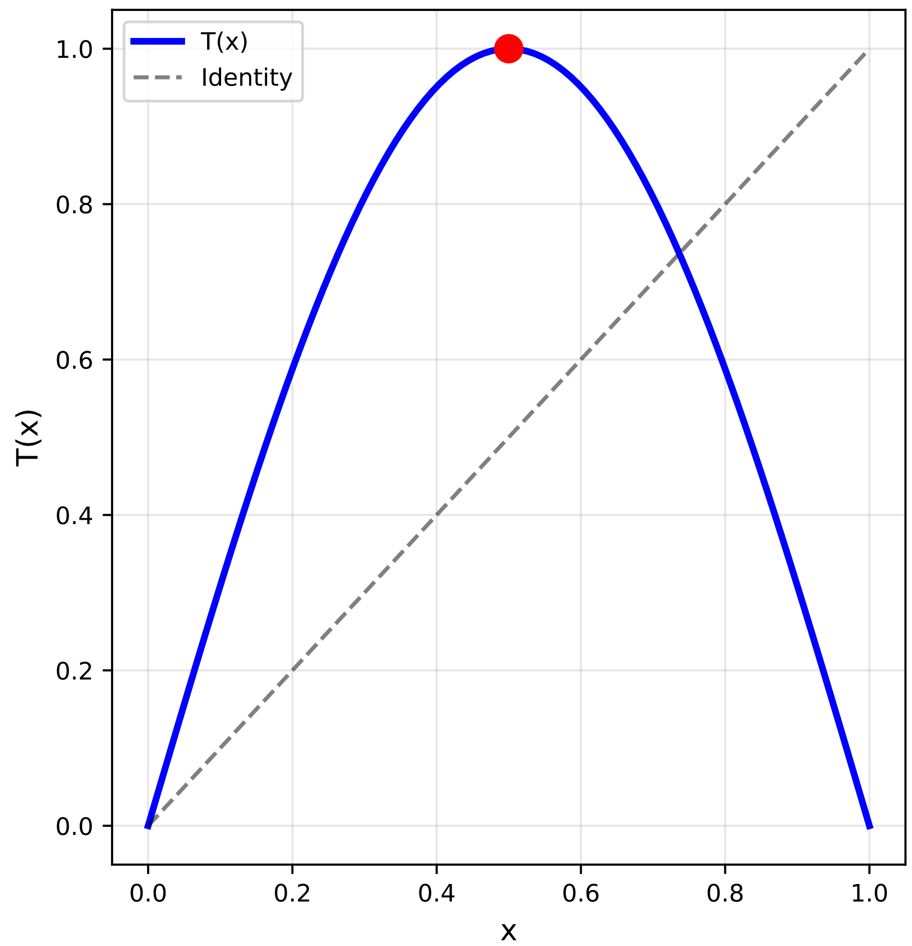
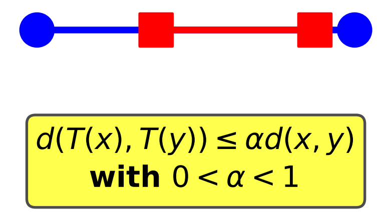
(1) Complete Metric Space (X, d)



(2) Continuous Mapping $T: X \to X$



(3) Contraction Property



BANACH CONTRACTION MAPPING THEOREM (Banach Fixed Point Theorem)

Let (X, d) be a complete metric space and $T: X \to X$ be a contraction mapping with Lipschitz constant $\alpha \in (0, 1)$, i.e.,

$$d(T(x), T(y)) \leq \alpha\, d(x, y) \text{ for all } x, y \in X$$

Then:

1. T has a unique fixed point $x^* \in X$
2. For any $x_0 \in X$, the sequence $\{x_n\}$ defined by $x_{n+1} = T(x_n)$ converges to x^*
3. The rate of convergence is exponential: $d(x_n, x^*) \leq \alpha^n d(x_0, x^*)$